

UNF

UNF is the beginning process for database normalization, it is composed of the list of attributes with the indication of repeating data for group of attributes.

Here, the repeating group is the collection of related attributes whose values can be repeated multiple times.

All the attributes with their respective entities are listed out the repeating group are indicated using curly braces.

Employee (organization_id, organization_name, { department_id, department_name, { manager_employee_id, manager_employee_name, { client_employee_id, first_name, middle_name, last_name, preferred_name, job_code, job_title, job_start_date, dob, hire_date, recent_hire_date, anniversary_date, term_date, years_of_experience, work_email, address, city, state, zip, country, fte_status, is_per_diem, cell_phone, work_phone, scheduled_weekly_hour, active_status, termination_reason, clinical_level } } })

Here the employee, department and manager group are repeating group and are indicated using {} bracket.

Organization can have multiple departments, department can have multiple manager, manager have multiple employees under him/her

1NF

The entity is in first normal form if it doesn't contain any repeating groups and single value or each row properties are held. In first normal form, the repeating group are separated into a new entity, the unique key for each new entity is identified and relationship between the two or more tables are created by addition of the foreign keys.

For Employee Entity

Employee (client_employee_id (PK), first_name, middle_name, last_name, preferred_name, job_code, job_title, job_start_date, dob, hire_date, recent_hire_date, anniversary_date, term_date, years_of_experience, work_email, address, city, state, zip, country, fte_status, is_per_deim, cell_phone, work_phone, scheduled_weekly_hour, active_status, termination_reason, clinical_level, organization_id (FK), department_id (FK), manager_employee_id (FK))

Primary Key: client_employee_id **foreign Keys:** organization_id, department_id, manager_employee_id

Organization (organization_id (PK), organization_name)

Primary key: Organization_id

Department (Department_id (PK), organization_id (FK), Department_name)

Primary key: Department_id **foreign key:** organization_id

Manager (manager_employee_id (PK), department_id (FK), organization_id (FK), manager_employee_name)

Primary key: manager_employee_id, **foreign key:** department_id , organization_id

2NF

In second normal form the partial dependencies between the non-prime attribute of composite key or primary key if there is only one unique identifier is removed by separating into a new entity and new relation is created.

For employee entity

client_employee_id $\longrightarrow$ first_name, middle_name, last_name, preferred_name, job_code, job_title, job_start_date, dob, hire_date, recent_hire_date, anniversary_date, term_date, years_of_experience, work_email, address, city, state, zip, country, fte_status, is_per_deim, cell_phone, work_phone, scheduled_weekly_hour, active_status, termination_reason, clinical_level

For organization entity

Organization $\longrightarrow$ Organization Name

For Department Entity

Department_id $\longrightarrow$ Department_name

For Manger Entity

manager_employee_id $\longrightarrow$ manager_employee_name

Since all the non-prime attributes are fully dependent on the primary key, it is in 2nf.

3NF

In 3nf the transitive dependency is removed, where the transitive dependency indicates that the non-prime attributes depend on the other non-prime attributes rather than the primary key of the table.

In case of **Employee entity**, job title depends on the job code, which violates the 3nf rules

client_employee_id -> job_code

Job_title -> job code

New entity **employee_job** is created

Employee_job (job_code (PK), job_title)

In case of city, state, country it depends on zip code, but through the data inspection zip code column value is found to be null, also as per the logic city depends on state, state depends on country, the over normalization process can create complex structure.

```
df[['zip']].isna().sum()
✓ 0.0s
zip    50
dtype: int64
```

Final table:

Employee (client_employee_id (PK), first_name, middle_name, last_name, preferred_name, job_code(FK), job_start_date, dob, hire_date, recent_hire_date, anniversary_date, term_date, years_of_experience, work_email, address, city, state, zip, country, fte_status, is_per_deim, cell_phone, work_phone, scheduled_weekly_hour, active_status, termination_reason, clinical_level, organization_id (FK), department_id (FK), manager_employee_id (FK))

Organization (organization_id (PK), organization_name)

Department (Department_id (PK), organization_id (FK), Department_name)

Manager (manager_employee_id (PK), department_id (FK), organization_id (FK), manager_employee_name)

Employee_job (job_code (PK), job_title)

